

Khawaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Mid-term Examination: Spring – 2022

Program: B.Sc. in CSE (Batch – 12th) 1st Year 2nd Semester

Course Title: Physics-II

Course Code: PHY 1201

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

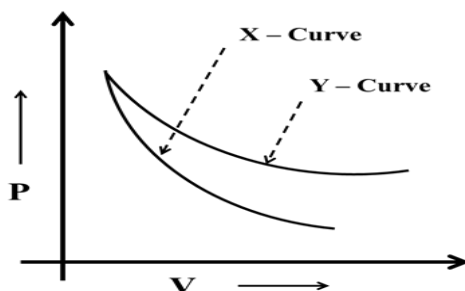
- What is the value of Rydberg constant?
 - $910,678 \text{ cm}^{-1}$
 - $678,910 \text{ cm}^{-1}$
 - $678,109 \text{ cm}^{-1}$
 - $109,678 \text{ cm}^{-1}$
 - $109,876 \text{ cm}^{-1}$
- Angular momentum of an electron in the 1st orbit of atom is equal to
 - $h/2\pi$.
 - h/π .
 - $2h/\pi$.
 - $nh/2\pi$.
 - nh/π .
- 'Production of heat by friction' is the example of process.
 - reversible
 - irreversible
 - adiabatic
 - isothermal
 - isobaric
- "Internal energy depends only on the temperature but not on pressure or volume" – stated by
 - Mayer.
 - Joule.
 - Van der Waal's.
 - Sadi Carno.
 - Brown.
- Which one is the correct relation between C_p & C_v ?
 - $C_p < C_v$
 - $C_p = C_v$
 - $C_p > C_v$
 - $C_p + C_v = R$
 - $C_v - C_p = R$
- When heat is supplied to a gas of volume 100m^3 at normal pressure, the volume changes to 100.2 m^3 . How much work done by the gas?
 - 40460J
 - 30360J
 - 20200J
 - 10160J
 - 0.0060J

7. For adiabatic process – (Where, P -Pressure, V -Volume, S -Entropy, T -temperature, W -external work done)
- a) $\Delta P = 0$
 - b) $\Delta V = 0$
 - c) $\Delta S = 0$
 - d) $\Delta T = 0$
 - e) $\Delta W = 0$
8. What is the unit of entropy?
- a) Jkg^{-1}
 - b) $calJ^{-1}$
 - c) $Jcal^{-1}$
 - d) JK^{-1}
 - e) KJ^{-1}
9. What is the value of γ for Helium gas?
- a) 1.33
 - b) 1.40
 - c) 1.50
 - d) 1.60
 - e) 1.66
10. What is the energy of the 1st orbit of hydrogen atom?
- a) -17.6 eV
 - b) -13.6 eV
 - c) -1.5 eV
 - d) -0.53 eV
 - e) 0.53 eV

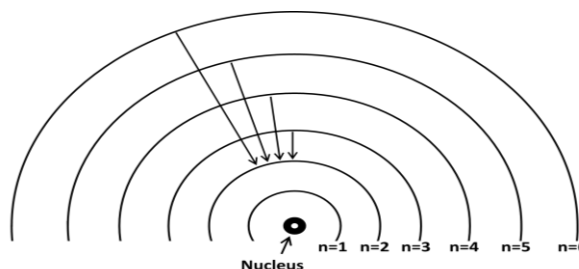
(Answer Question No. 1 and any three from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1. See the figure for thermodynamic system and answer the questions that follows:



- a) Specify the X – Process. [1]
 - b) Identify which curve follows the Boyle's law. [1]
 - c) Is it possible to establish a relation of slope of above two curves? – Justify your answer. [3]
- a) Disclose a condition of elastic collision. [1]
 - b) What are the factors on which the mean free path of a gas depends? [2]
 - c) The mean free path of the molecules of a gas is $6 \times 10^{-8}\text{m}$ and the diameter of a molecule is $2.5 \times 10^{-10}\text{m}$. Find the number of molecules per m^3 . [2]
- a) What are the requirements to get 100% efficient engine? Are these possible? [1]
 - b) Imply the total work done of Carnot's engine in Carnot's cycle with the diagram. [2]
 - c) Comments about the far future of universe related to entropy with logical explanation. [2]
- a) State the Zeroth law of thermodynamics. [1]
 - b) Clarify the significance of the positive and negative value of Gibb's free energy in the reversible reaction of an electrochemical cell. [2]
 - c) Calculate the work done in isochoric process. [2]
5. See the spectral series of hydrogen atom and give answer of questions that follows:



- a) Give the name of the series of above figure. [1]
- b) Find the wave number for that. [2]
- c) Draw others possible spectral series for above mentioned atom. [2]